A Bayesian WeightWatcher: Calibrated Heavy-Tailed Spectral Diagnostics and What They Overturn

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Paper under double-blind review

Abstract

Heavy-Tailed Self-Regularization (HTSR) reads the quality of a trained neural network off the eigenvalue spectrum of its weight matrices: the power-law tail exponent α is a label-free generalization proxy, and Martin's renormalization-group theory sharpens this into the prediction that $\alpha = 2$ is a robustness-optimal fixed point. The reference tool, WeightWatcher, reports α as a single point estimate from one Kolmogorov–Smirnov scaling window, and never tests whether a layer is a power law at all. We show this matters. We introduce wwj/wwjd, a JAX/Equinox reimplementation that adds a calibrated Bayesian layer: a closed-form Gamma–Pareto posterior over α, Bayesian model averaging over the scaling window, Bayesian model comparison against exponential, lognormal, truncated power-law, and generalized-Pareto tails, posterior-predictive checks, a hierarchical cross-layer posterior, and an errors-in-variables regression that turns α into a calibrated accuracy predictor. Reanalyzing WeightWatcher's own canonical checkpoints, we find: the famous "GPT2 is better trained than GPT" gap largely dissolves once the scaling window is marginalized (mean α 4.12 $\to$ 2.90 for GPT, the GPT–GPT2 gap shrinking threefold), with two independent goodness-of-fit tests contradicting the "not a power law" mechanism and showing the inflation is truncated-power-law misspecification; the flagship VGG accuracy trend is confirmed and sharpened under the posterior ($r{=}-0.77 \to -0.98$); and batch normalization breaks the power law in 8–17% of layers, invisible to the point estimate. We then apply the same diagnostics across modalities, where the $\alpha = 2$ robustness prediction holds for physics-realistic neuroimaging perturbations but fails to transfer to a foundation-model-scale pathology intervention. We release wwj as an open tool for uncertainty-aware spectral diagnostics.

1 Introduction

The Heavy-Tailed Self-Regularization (HTSR) framework (Martin & Mahoney, 2021a; 2019) observes that the empirical spectral density (ESD) of weight matrices in well-trained deep networks develops a heavy, approximately power-law tail $p(\lambda) \propto \lambda^{-\alpha}$, and that the exponent α predicts generalization across hundreds of pretrained models with no access to data or labels (Martin et al., 2021). Martin's renormalization-group (RG) theory of learning (Martin, 2026) and its semi-empirical successor (Martin & Mahoney, 2025) sharpen this into a quantitative claim: $\alpha = 2$ is a critical fixed point, and proximity to it should confer robustness.

These are strong, falsifiable claims, and they rest entirely on how α is estimated. The reference tool, WeightWatcher, fits α by maximum likelihood on the tail above a single scaling-window lower bound $x_{\min}$ chosen to minimize a Kolmogorov–Smirnov (KS) distance, and reports a per-layer point value. This pipeline has two well-known soft spots that the authors themselves flag: the MLE "works very well for $\alpha \in (2, 4)$...adequate, although imprecise, for smaller and especially larger α" (Martin et al., 2021), and the KS objective sometimes has no well-defined minimum in $x_{\min}$ (Martin & Mahoney, 2021b); the tool ships a truncated-power-law option precisely because plain power-law fits over-estimate α. Yet downstream conclusions—"this model is poorly trained because it has unusually large α," "this layer is ideal because $\alpha \approx 2$"—are read off the point estimate as if it carried no uncertainty, and the power-law assumption underlying α is never tested.

We argue that a Bayesian treatment is not a cosmetic addition but changes published conclusions. We introduce wwj, a JAX/Equinox (Bradbury et al., 2018; Kidger & Garcia, 2021) reimplementation of WeightWatcher, and wwjd, a Bayesian layer on top of it (Section 3): a closed-form conjugate posterior over α, Bayesian model averaging (BMA) over $x_{\min}$, Bayesian model comparison that adds the truncated power law and the extreme-value generalized Pareto to the usual exponential/lognormal alternatives, posterior-predictive goodness-of-fit, a hierarchical cross-layer population posterior, and an errors-in-variables (EIV) regression that propagates α uncertainty into a calibrated accuracy predictor, together with region-of-practical-equivalence and prior-sensitivity diagnostics.

Our contributions are: (1) the wwj/wwjd tool, including exact CSN $x_{\min}$ selection (Clauset et al., 2009), a differentiable Hill estimator enabling direct spectral regularization, and the Bayesian model-criticism suite; (2) a reanalysis of WeightWatcher's own canonical checkpoints (Section 4) showing that the GPT/GPT2 "training quality" gap is largely a scaling-window artifact, that the flagship VGG accuracy trend strengthens under the posterior, and that batch normalization silently breaks the power law; and (3) a cross-modality application (Section 5) where the $\alpha = 2$ robustness prediction holds for realistic neuroimaging perturbations but does not transfer to a pathology foundation-model intervention—an honest map of where the theory does and does not pay off.

2 Related Work

HTSR and spectral quality metrics. Martin & Mahoney (2021a; 2019) established heavy-tailed weight spectra and the "5+1 phases" of training; Martin et al. (2021) showed α predicts test-accuracy trends without data. The capacity metric is the weighted alpha $\hat{\alpha} = \mathrm{mean}_\ell \alpha_\ell \log_{10} \lambda_{\max,\ell}$. The RG grounding is Martin (2026); Martin & Mahoney (2025); the contest post-mortem (Martin & Mahoney, 2021b) documents a Simpson's paradox between shape (α) and scale metrics. All of these read α as a point estimate; we put a posterior on it.

Power-law fitting and its critique. The CSN MLE-plus-KS procedure (Clauset et al., 2009) and the powerlaw package (Alstott et al., 2014) underlie HTSR fitting. The statistical risks—$x_{\min}$ instability, alternative heavy-tailed distributions, finite-size truncation—are well documented, and motivate Bayesian model averaging over $x_{\min}$, model comparison, and the truncated power law and generalized Pareto from extreme-value theory (Coles, 2001) that we add.

Bayesian workflow and model criticism. Power-scaling prior-sensitivity (Kallioinen et al., 2024) and predictive model comparison (Vehtari et al., 2017) are standard tools for trustworthy Bayesian conclusions; we adapt them to spectral diagnostics.

Spectral shaping and robustness. The Muon optimizer (Jordan et al., 2024) is conjectured to shape spectra toward the RG optimum; our differentiable alpha_loss is a direct, optimizer-agnostic alternative. For the cross-modality application we use TorchIO (Pérez-García et al., 2021) perturbations, the MONAI (Cardoso et al., 2022) ViT, and the BrainIAC (Pati et al., 2024) and MindEye (Scotti et al., 2024) checkpoints.

3 Method: wwj and wwjd

Frequentist core (wwj). For each 2D weight matrix W with min(shape) ≥ 50 we compute the eigenvalues of $W^\top W/N$ and select $x_{\min}$ by exhaustive KS minimization over every eigenvalue as a candidate (matching CSN 2009, rather than the reference's log-spaced grid). A differentiable Hill estimator $\hat{\alpha} = 1+n(\sum_i \log \lambda_i/x_{\min})^{-1}$ backpropagates through eigvalsh, enabling the regularizer $\mathcal{L}_\alpha = \lambda_{\text{reg}} \cdot \mathrm{mean}_\ell(\hat{\alpha}_\ell - 2)^2$. A Vuong likelihood-ratio test compares power-law against exponential and lognormal tails.

Bayesian layer (wwjd). The tail above $x_{\min}$ is Pareto, so $t = \log(\lambda/x_{\min})$ is exponential with rate $\beta = \alpha - 1$, and a conjugate $\mathrm{Gamma}(a_0, b_0)$ prior on β yields the closed-form posterior $\beta \mid \text{data} \sim \mathrm{Gamma}(a_0 + n, b_0 + \sum t)$. Hence the posterior mean, credible interval, and $P(\alpha < 2)$ are exact, and the flat-prior limit recovers the CSN MLE. On top of this single-window posterior we build: (i) BMA over $x_{\min}$, a Gamma mixture weighting each candidate window by a comparable marginal likelihood, with a law-of-total-variance decomposition that separates within-window from window-choice uncertainty; (ii) Bayesian model comparison with analytic evidences for power-law/exponential/lognormal and Laplace evidences for the truncated power law and generalized Pareto; (iii) a posterior-predictive p-value from Lomax replication; (iv) a hierarchical NumPyro population posterior over per-layer α with shrinkage; (v) an errors-in-variables regression of a downstream metric on $\hat{\alpha}$ that propagates the $\hat{\alpha}$ posterior SD as measurement error; and (vi) ROPE ($P(\alpha \in [2-\delta, 2+\delta])$) and power-scaling prior-sensitivity diagnostics. The conjugate per-layer surfaces are pure JAX; only the hierarchical model samples.

4 Validation on WeightWatcher's Own Canonical Cases

Before applying the Bayesian diagnostics to new modalities, we test them where the answer is already published: on the exact pretrained checkpoints Martin & Mahoney used to establish heavy-tailed self-regularization. We load each model from its public source (torchvision ImageNet weights; HuggingFace GPT/GPT2), extract every 2D weight matrix with $\min(\text{shape}) \geq 50$, and compute both the frequentist WeightWatcher point estimate and the wwjd posterior. Three findings result.

The VGG weighted-α trend is confirmed—and sharper under the posterior. WeightWatcher's flagship claim is that the weighted-alpha $\hat{\alpha} = \mathrm{mean}_\ell\, \alpha_\ell \log_{10} \lambda_{\max,\ell}$ tracks test accuracy across the VGG11/13/16/19 (±BN) ImageNet series. We reproduce this with the frequentist point estimate (Pearson $r = -0.77$ between $\hat{\alpha}$ and top-1 accuracy) and find the Bayesian $\hat{\alpha}$ (posterior-mean per layer) tracks accuracy more strongly, $r = -0.98$. The Bayesian model-averaging over $x_{\min}$ pulls down the inflated large-α layers (see below), making the metric more predictive, not less. We note that the metric must be the per-layer mean: the unweighted mean α gives $r = +0.89$ (the wrong sign—a depth-confounded Simpson's paradox), which the weighting corrects, and which the posterior recovers cleanly.

We close the loop with an errors-in-variables regression of accuracy on $\hat{\alpha}$ that propagates the per-model $\hat{\alpha}$ posterior uncertainty (Section 3). The posterior slope is -6.2 percentage points of top-1 accuracy per unit $\hat{\alpha}$ (95% credible interval $[-7.8, -4.8]$, $P(\text{slope} < 0) = 1.00$), with residual $\sigma = 0.31$pp: the weights-only spectrum predicts ImageNet top-1 to within a third of a point. The measurement-error slope is slightly steeper than the naive ordinary-least-squares slope (-6.0), the expected attenuation correction. This turns WeightWatcher's reported correlation into a calibrated predictor with honest prediction intervals.

The "GPT2 is better-trained than GPT" gap largely dissolves under $x_{\min}$ marginalization. The canonical NLP example holds that GPT is poorly trained because it has "numerous unusually large α" layers, while GPT2-small has all $\alpha \leq 6$. Under the frequentist single-$x_{\min}$ estimate we reproduce this: mean $\alpha = 4.12$ for GPT versus 3.62 for GPT2 and 3.70 for GPT2-medium. Under the wwjd posterior that marginalizes over the scaling window, all three collapse into one healthy band: 2.90, 2.73, 2.75 respectively. The frequentist gap (GPT $-$ GPT2 $= 0.50$) shrinks roughly threefold to 0.16. Per layer, GPT's nine 768×768 attention-projection matrices fall from frequentist $\alpha \in [5.2, 6.9]$ to posterior $\alpha \in [2.8, 3.5]$. The shift is not an artifact of our estimator: 18–36% of the per-layer α variance on these layers is attributable to the $x_{\min}$-window choice alone (the law-of-total-variance decomposition of the posterior), exactly the degree of freedom the single KS-optimal window collapses.

Two independent goodness-of-fit tests then contradict the "not well-described by a power law" mechanism on these same layers. The Bayesian model comparison prefers a power law over both exponential (log Bayes factor +120 to +163) and lognormal (+24 to +35) tails, and the posterior-predictive p-values are all comfortably above 0.05 (0.26–0.68): the power-law fit is adequate in absolute terms. When we extend the comparison set to the

Table 1: Bayesian reanalysis of WeightWatcher's canonical checkpoints. $\hat{\alpha}/\alpha$ columns are weighted-/mean-alpha; "freq" is the single-$x_{\min}$ WeightWatcher estimate, "bayes" the wwjd posterior (BMA over $x_{\min}$). PL-rej. = fraction of layers the Bayes factor rejects as non-power-law.

Model	mean α (freq)	mean α (bayes)	PL-rej.	note
openai-gpt	4.12	2.90	0%	"poorly trained" (freq) → healthy (bayes)
gpt2	3.62	2.73	0%	gap to GPT shrinks 3×
gpt2-medium	3.70	2.75	0%	
vgg19	2.71	2.42	0%	$\hat{\alpha}$ vs acc: r_{freq}=−0.77
vgg19_bn	2.93	2.42	17%	$\hat{\alpha}$ vs acc: r_{bayes}=−0.98

truncated power law (the model WeightWatcher itself ships because plain power-law fits over-estimate α) and the generalized Pareto (the extreme-value-theory peaks-over-threshold tail), the truncated power law beats the plain one on all 50 GPT layers (mean log Bayes factor −1.54). The "large α" is therefore a power-law-with-truncation mis-specification, made explicit: when the finite-size cutoff is modeled, the inflation disappears. A region-of-practical-equivalence test keeps this honest—only 1/50 GPT layers place majority posterior mass in $\alpha \in [1.75, 2.25]$, so the Bayesian $\alpha \approx 2.9$ is far below the frequentist 4.1 but still above the RG-ideal 2.0. A power-scaling prior-sensitivity diagnostic confirms the conclusions are not prior-driven (mean sensitivity 0.001 posterior-SD units).

Batch normalization breaks the power law in a way the point estimate hides. Across the BatchNorm VGG variants, the Bayesian model comparison rejects the power law (in favor of exponential or lognormal) for 8–17% of layers, rising with depth (vgg19_bn: 17%); plain VGG and all GPT models reject for 0%. The frequentist α, which never tests the power-law assumption, is blind to this structure.

Table 1 summarizes. The unifying mechanism across both image and language models is that the frequentist single-window estimate inflates α on hard-to-fit layers, and the larger the frequentist α, the larger the posterior correction; clean low-α layers barely move. This is the over-estimation Martin acknowledges as the reason for the truncated-power-law option—here quantified, calibrated, and shown to change a published conclusion.

5 Cross-Modality Application: Where $\alpha = 2$ Pays Off, and Where It Does Not

The reanalysis above validates the diagnostics on models with known answers. We now turn them on the open question: does proximity to $\alpha = 2$ actually confer robustness, across modalities? We summarize a positive neuroimaging result and a negative foundation-model-scale pathology result.

Neuroimaging (positive, partial). A cross-modality survey of five neuroimaging checkpoints places the BrainIAC lineage near the RG optimum (mean $|\alpha - 2| \in \{0.12, 0.40\}$) and the MindEye lineage far from it ($\{3.49, 4.73\}$); the Vuong/Bayesian comparison formally rejects the power law for the MindEye per-subject layers with $\alpha \in [8, 11]$, a silent failure of bare α reporting. In a paired cohort sweep over 20 WAND subjects under TorchIO perturbations, the spectrally-closer checkpoint shows 3× lower CLS-embedding drift under bias field (Wilcoxon $p = 0.00022$, 9/10 perturbation seeds $p < 0.05$), with the effect concentrated in smooth, physics-realistic perturbations (bias field, motion) and absent for high-frequency artifacts (spike, noise)—the pattern the RG theory predicts. A controlled MLP ablation confirms the causal direction: alpha_loss drives mean α to 2.00 within 1000 steps and uniformly improves noise robustness.

Pathology (negative). The same alpha_loss regularizer, applied at foundation-model scale during continual DINOv2 pretraining of a ViT-S pathology encoder (10^{18}-FLOP budget), is robustness-neutral: pulling backbone weight-matrix α toward 2 leaves the downstream robustness probe unchanged and the mean probe score statistically indistinguishable from the unregularized recipe. The MLP-scale robustness gain does not transfer to a real foundation model in this modality. We report this as an honest boundary on the $\alpha = 2$ prediction: the diagnostic value of α (and of the Bayesian corrections above) is robust, but the interventional robustness payoff is scale- and modality-dependent.

6 Discussion

A single thread runs through the reanalysis: the frequentist single-window estimate inflates α on hard-to-fit layers, and the larger the frequentist α, the larger the posterior correction, because those are exactly the layers where the scaling window is ambiguous (a large share of their α variance is the window choice) and where the plain power law is mis-specified relative to a truncated one. Clean, near-optimal layers barely move. This is why a published conclusion built on large point-α layers—"GPT is poorly trained"—can weaken or dissolve under marginalization, while a conclusion built on the depth trend of well-behaved layers—the VGG accuracy relationship—survives and even sharpens. The practical recommendation is to report α with a credible interval, a power-law validity check, and, where the conclusion is comparative, the window-choice variance share; wwjd makes all three one function call.

Limitations. The Laplace evidences for the truncated power law and generalized Pareto are approximations; the cross-modality negative result is a single intervention; and the neuroimaging positive result, while controlled, spans one architecture family. Predictive model comparison (PSIS-LOO) and a mixture model separating the Marchenko–Pastur bulk from the heavy tail are natural next additions.

7 Conclusion

Heavy-tailed self-regularization is a powerful, falsifiable theory, and exactly because it is falsifiable it deserves uncertainty-aware estimation. We showed that a calibrated Bayesian treatment of the power-law exponent—posteriors, window marginalization, model comparison, and goodness-of-fit—changes conclusions drawn from the reference tool on its own canonical examples, turns the quality metric into a calibrated accuracy predictor, and draws an honest map of where the $\alpha = 2$ robustness prediction transfers across modalities. We release wwj/wwjd so that spectral diagnostics can be reported with error bars.

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Supplementary Material